



Greater Noida

Report

Of

**C++ PROGRAMMING INTERNSHIP:
CONSOLE APPLICATIONS, OBJECT-ORIENTED
PROGRAMMING AND FILE HANDLING**

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 25SCS1003005205

Name of Student: RAVI SHANKAR KUMAR

Semester: 3rd Semester

CANDIDATE'S DECLARATION

I, RAVI SHANKAR KUMAR, Roll No. 25SCS1003005205, do hereby declare that the internship report titled "C++ Programming Internship: Console Applications, Object-Oriented Programming and File Handling" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: RAVI SHANKAR KUMAR

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Thiranex – Skill Development & Future Tech for providing me with the opportunity to undergo this C++ Programming Internship, and for the structured tasks, mentorship and resources that helped me build the projects discussed in this report. I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout this internship.

I express my gratitude to my parents for their blessings.

Signature

Student Name: RAVI SHANKAR KUMAR

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Internship Completion Certificate

The internship offer letter and the certificate of completion issued by Thiranex – Skill Development & Future Tech on successful completion of the C++ Programming Internship are attached below.

Annexure I: Internship Offer Letter



Date: 30 July 2026

TO,

Ravi shankar kumar

Intern ID: THX-JUL3026-505

Subject: Offer Letter for Internship in C++ Programming

Dear **Ravi shankar kumar**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - C++ Programming
Commencement Date	30 Jul 2026
Completion Date	29 Aug 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



Annexure II: Certificate of Internship Completion



4. Project Description

4.1 Introduction

This report documents the work undertaken during a C++ Programming Internship, completed with Thiranex – Skill Development & Future Tech, as part of the academic requirement for the 3rd semester B.Tech CSE curriculum at IILM University, Greater Noida. The internship combined structured, milestone-based assignments with hands-on, project-based learning, and ran from 30 July 2026 to 29 August 2026 (approximately 4.5 weeks) in Remote / Project-Based mode (Intern ID: THX-JUL3026-505).

As part of the practical component, four progressive modules were completed, each building on core C++ programming concepts: (i) a console-based Student Management System using structures and file handling, (ii) a Bank Management Application built using object-oriented programming, (iii) a Library Management System combining structured records with search functionality, and (iv) an interactive Mini Game Project (Tic Tac Toe / Snake Game) demonstrating loops, arrays and conditional logic. Together, these modules trace a practical progression through core C++ programming, object-oriented design, file handling and interactive program logic. Each module is described in detail in Section 4.8 (Methodology), and a module-wise summary of the internship curriculum is presented in Section 4.10 (Internship Training Log Book).

4.2 Organization Profile

Thiranex – Skill Development & Future Tech is a skill-development and future-technology focused platform that offers structured, project-based internship programs across domains including C++ programming, web development and future technologies. The organization's programs are designed to build industry-relevant, hands-on competencies through progressively advanced, real-world coding assignments, with continuous online progress tracking and certification on successful completion. The internship undertaken for this report was carried out entirely in Remote / Project-Based mode and ran from 30 July 2026 to 29 August 2026.

4.3 Problem Statement

The internship's practical component was centred on building four independent, increasingly involved C++ console applications, with each stage addressing a distinct programming problem:

- **Persistent Record Management:** Given the need to manage student records reliably, the task was to build a console-based Student Management System combining structures/classes with file handling for persistent storage.
- **Object-Oriented System Design:** Given the need to simulate banking operations, the task was to design a Bank Management Application using object-oriented programming concepts (classes and objects) with secure, file-based data storage.
- **Structured Data & Search:** Given the need to track books, members and borrowing activity, the task was to build a Library Management System supporting issue/return workflows and search by title or author.
- **Interactive Program Logic:** Given the need to demonstrate core programming constructs interactively, the task was to build a console-based mini game (Tic Tac Toe / Snake) using loops, arrays and conditional logic.

Together, these problems are representative of the standard progression of C++ application development: structures and functions, object-oriented design, file-based persistence, and interactive program logic.

4.4 Project Objectives

- To master core C++ concepts — structures, classes, functions, loops and arrays — and apply OOP principles to real-world style problems.
- To implement reliable file handling (fstream) so that records persist correctly across program runs.
- To design clean, menu-driven console interfaces with robust input validation.
- To implement search and record-management functionality (issue/return, search-by-field) for structured data.
- To apply programming logic — loops, arrays and conditions — to build interactive, rule-based programs.

4.5 Scope of the Project

The scope of the internship is limited to educational, console-based C++ implementations using standard file handling for persistence, and is not intended to represent a full production software system with a networked back-end or database layer. The Student Management System, Bank Management Application, Library Management System and Mini Game Project are each scoped as independent, self-contained console applications, built and tested individually. The focus throughout was on correctly applying core C++ programming, object-oriented design and file-handling fundamentals rather than on building a large-scale, multi-service system.

4.6 Technologies and Tools Used

- Programming Language: C++ (core syntax, control structures, functions)
- Object-Oriented Programming: Classes, objects, encapsulation
- File Handling: fstream-based read/write for persistent data storage
- Data Structures: Structures and arrays for structured, in-memory data management
- STL Concepts: Vectors, strings and standard container usage where applicable
- Development Environment: GCC / Code::Blocks / Dev-C++ (console-based compilation and execution)

4.7 System Architecture

The overall internship followed a layered console-application architecture: structured data models (structures/classes) form the foundation, over which core program logic (menu-driven operations, banking/library workflows, or game rules) is implemented; file handling is then used to persist data across sessions; and each application is validated and tested before being marked complete.

Layer 1	Data Modelling (Structures / Classes)
Layer 2	Core Program Logic (Functions)

Layer 3	Menu-Driven Console Interface
Layer 4	File Handling (Persistence)
Layer 5	Input Validation & Testing
Layer 6	Module Completion & Review

Fig. 4.1: Layered Console-Application Architecture Followed Across the Internship

4.8 Methodology

This section describes, module by module, the methodology followed to complete each of the four internship assignments.

4.8.1 Module 1: Student Management System

Due Date: 06 Aug 2026 | Status: Completed Successfully

Developed a console-based Student Management System in C++ using structures/classes, file handling and menu-driven operations.

- Designing a Student record structure/class capturing fields such as roll number, name, and marks.
- Implementing Add, Update, Delete and Display operations over the in-memory collection of student records.
- Using file handling (fstream) to persist student records to disk, so that data survives across separate program runs.
- Building a menu-driven console interface using loops and switch-case logic for smooth navigation between operations.
- Validating user input (e.g., numeric fields, duplicate roll numbers) to prevent corrupted records.

Outcome: Built a functional console application that allows users to add, update, delete, and display student information with persistent data storage for reliable record management.

4.8.2 Module 2: Bank Management Application

Due Date: 13 Aug 2026 | Status: Completed Successfully

Designed a C++ banking application simulating core banking operations using object-oriented programming and file handling.

- Designing an Account class encapsulating customer details, account number and balance, following object-oriented design principles.
- Implementing core banking operations — deposit, withdrawal, and balance check — as member functions with appropriate validation (e.g., insufficient balance checks).
- Using file handling to securely store and retrieve customer account data between sessions.
- Building a menu-driven interface allowing the user to select and perform banking operations on a chosen account.

- Testing the application against edge cases such as invalid account numbers and overdraft attempts.

Outcome: Delivered a secure and functional banking system capable of performing deposits, withdrawals, and balance inquiries while maintaining persistent customer records.

4.8.3 Module 3: Library Management System

Due Date: 20 Aug 2026 | Status: Completed Successfully

Developed a Library Management System to track books, members and borrowing activity.

- Implementing structures/classes to model Book and Member details (e.g., title, author, availability status, member ID).
- Building book issue and return features that update book availability and maintain a record of borrowing activity.
- Implementing search functionality allowing lookup of books by title or author.
- Using file handling to persist the book catalogue and membership/borrowing records across sessions.
- Designing a menu-driven interface to tie together book addition, issue, return and search operations.

Outcome: Delivered a functional application that enables book addition, issue and return processing, and search by title or author for streamlined library operations.

4.8.4 Module 4: Mini Game Project (Tic Tac Toe / Snake Game)

Due Date: 27 Aug 2026 | Status: Completed Successfully

Created a console-based mini game demonstrating loops, arrays and conditional logic.

- Representing the game board (Tic Tac Toe grid / Snake game state) using arrays, and rendering it to the console after every move.
- Implementing turn-based or real-time game logic using loops and conditional statements to enforce game rules.
- Adding win/loss (and draw, where applicable) detection logic evaluated after each move or game tick.
- Implementing a replay option allowing the user to restart the game without relaunching the program.
- Refining the code structure for readability, separating board-rendering, input-handling and game-logic responsibilities.

Outcome: Delivered an interactive game with dynamic board display, win/loss detection, and replay functionality, showcasing effective implementation of game logic.

4.9 Expected Outcomes

- A functional Student Management System supporting add, update, delete and display operations with persistent, file-based storage.
- A secure, object-oriented Bank Management Application capable of performing deposits, withdrawals and balance inquiries with persistent customer records.

- A functional Library Management System enabling book addition, issue/return processing, and search by title or author.
- An interactive console-based mini game with dynamic board display, win/loss detection and replay functionality.
- Practical, transferable familiarity with the complete C++ application-development workflow — data modelling, object-oriented design, file handling, and menu-driven interface design — using industry-standard tools.
- A foundation for further exploration of advanced data structures, algorithms, and larger, real-world C++ projects in future coursework.

4.10 Challenges Faced & Key Learnings

4.10.1 Managing File I/O Reliably

While building the Student Management System (Module 1), records were occasionally lost or corrupted when the program terminated unexpectedly mid-write. This was resolved by ensuring file streams were properly opened, checked, and closed after every read/write operation, and by handling file-open failures gracefully, leading to a much clearer understanding of safe, reliable file handling in C++.

4.10.2 Structuring Object-Oriented Designs

The first attempt at the Bank Management Application (Module 2) used loose, procedural functions operating on global data, which quickly became difficult to extend. Refactoring the design around an Account class with encapsulated data and member functions produced a much cleaner, more maintainable object-oriented structure.

4.10.3 Designing Robust Menu Systems

Early versions of the Library Management System (Module 3) crashed or misbehaved when users entered unexpected input at menu prompts. Adding thorough input validation and looping the menu until valid input was received made the console interface far more robust and user-friendly.

4.10.4 Implementing Game Logic Correctly

While building the Mini Game Project (Module 4), correctly detecting win/loss conditions and managing overall game state initially proved error-prone. Breaking the win-check logic into small, clearly-named helper functions and testing each condition independently resolved the issue and produced much more reliable game logic.

4.11 Skills & Competencies Gained

4.11.1 Core C++ Programming

- Structures, classes and object-oriented design principles.
- Control flow, loops and array-based data handling.
- Functions and modular code organization.

4.11.2 Data Persistence

- File handling with fstream (read/write) for persistent storage.
- Structuring records for reliable, file-based persistence.
- Input validation and defensive error handling.

4.11.3 Application Design

- Menu-driven console interface design.
- CRUD-style record-management workflows (add/update/delete/search).
- Interactive program/game logic, including win/loss detection and replay handling.

4.12 Internship Training Log Book

The internship curriculum was delivered through four structured, milestone-based modules, each with a defined due date, a set of key topics/features, and an expected outcome. Table 4.1 summarises the module-wise progress; all four milestones were completed successfully and on schedule.

Module	Module Title	Due Date	Key Topics Covered	Key Learning Outcome
1	Student Management System	06 Aug 2026	Structures/classes for student records; file handling (fstream) for persistent storage; menu-driven console navigation; input validation.	Built a functional console application that allows users to add, update, delete, and display student information with persistent, file-based storage.
2	Bank Management Application	13 Aug 2026	Object-oriented programming (classes & objects); encapsulation of account data; file handling for customer records; deposit/withdrawal/balance-check operations.	Delivered a secure, object-oriented banking system capable of performing deposits, withdrawals and balance inquiries while maintaining persistent customer records.
3	Library Management System	20 Aug 2026	Structures/classes for books and members; book issue/return workflows; search functionality; file-based persistence of library records.	Delivered a functional application enabling book addition, issue/return processing, and search by title or author for streamlined library operations.
4	Mini Game Project (Tic Tac Toe / Snake Game)	27 Aug 2026	2D arrays for game-board representation; loops and conditional logic for game rules; win/loss detection;	Delivered an interactive game with dynamic board display, win/loss detection and replay functionality,

Module	Module Title	Due Date	Key Topics Covered	Key Learning Outcome
			replay handling.	demonstrating effective game logic.

Table 4.1: Internship Training Log Book – Module-wise Summary

4.13 Certificates of Completion (Annexure at a Glance)

The Internship Offer Letter and the Certificate of Internship Completion issued by Thiranex – Skill Development & Future Tech are attached in full in the preceding section (Internship Completion Certificate) of this report as Annexure I and Annexure II respectively.



Table 4.2: Internship Offer Letter and Completion Certificate – Annexure Summary

Certificate ID	THX-JUL3026-505
Milestones Completed	4 / 4
On-Track Submissions	100%
C++ Applications Built	4

5. Bibliography / References

- [1] Stroustrup, B., "The C++ Programming Language," 4th ed., Addison-Wesley, 2013.
- [2] cplusplus.com, "C++ Language Reference," <https://cplusplus.com/reference/>
- [3] cppreference.com, "C++ Standard Library Reference," <https://en.cppreference.com/>
- [4] cplusplus.com, "Input/Output with files," <https://cplusplus.com/doc/tutorial/files/>
- [5] GeeksforGeeks, "Object Oriented Programming in C++," <https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/>
- [6] GeeksforGeeks, "File Handling in C++," <https://www.geeksforgeeks.org/file-handling-c-classes/>
- [7] GNU Project, "GCC, the GNU Compiler Collection — Documentation," <https://gcc.gnu.org/onlinedocs/>
- [8] Thiranex – Skill Development & Future Tech, Internship Program Materials, Task Briefs and Milestone Portal, C++ Programming Virtual Internship, 2026.